

Places to Visit Guide - Project Proposal (v2 - updated APIs)

Course: WDD330 · **Author:** Erick Scala

v2 replaces OpenStreetMap/Nominatim tiles + OpenTripMap with Geoapify (geocoding, key), Wikipedia (places + details, no key) and Esri (tiles, no key). Verified working Oct 2026.

1. General Description

Problem: When planning a trip, it can be difficult to discover what is near a destination, filter results by place type, and view locations on a map all in one place.

Solution: "Places to Visit Guide" is a web application where users enter a city or address (or use their current location) to get a list and an interactive map of nearby tourist attractions.

Motivation: I am passionate about traveling and discovering off-the-beaten-path locations, and I have always found it challenging to find-in a single place-what to visit at a destination and where those spots are located. Additionally, this project is an ideal opportunity to strengthen a skill I have not yet mastered: consuming REST APIs.

2. Target Audience

- Travelers and tourists wishing to discover places to visit in a new destination.
- Families and groups planning weekend outings in their local area.
- Students and budget-conscious travelers looking for free options that do not require creating an account.
- Mobile users checking the guide while exploring the city.

3. Major Functions

N°	Feature	Description
1	Location search	Search field for a city, country, or address. Uses Geoapify Geocoding (<code>/v1/geocode/search?text=...&apiKey=...</code>) to convert the text into coordinates (the first result is used). The search runs when the user presses Enter or the search button.
2	Interactive map	Map (Leaflet + Esri World Street Map tiles, URL order <code>{z}/{y}/{x}</code>) centered on the destination, with a marker for each point of interest and the required <code>© Esri</code> attribution credit.
3	Attractions list	Cards showing name, category, distance to the center (computed locally with haversine), and popularity. Data from Wikipedia Geosearch (<code>?list=geosearch</code>). Clicking a card highlights its corresponding marker.

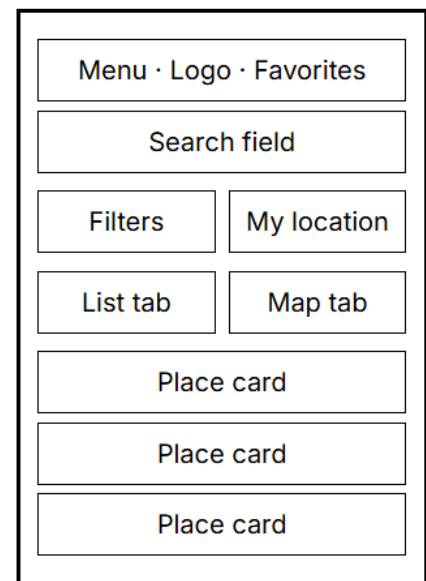
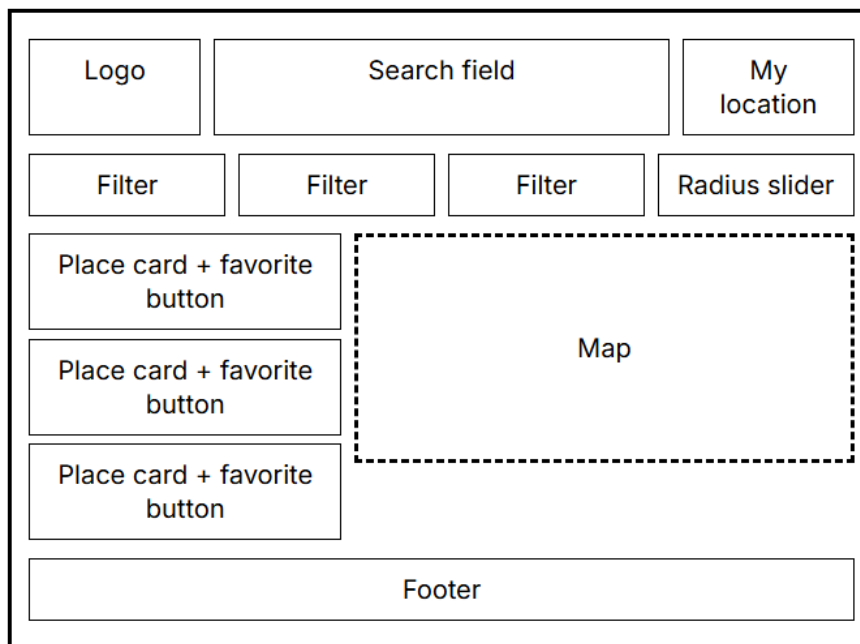
N°	Feature	Description
4	Category filter	Buttons/chips to filter by: museums, architecture, nature, religion, history, entertainment (mapped to Wikipedia search terms/categories client-side).
5	Place details	Modal view or page featuring an image (`pageimages`), description (`extracts`), coordinates, Wikipedia link, and a "get directions" button (Google Maps/Esri directions link). Data from Wikipedia API (`prop=pageimages`)
6	Favorites	Save/remove places using a heart icon. Stored in localStorage with a dedicated view.
7	Use my location	Button using the browser's Geolocation API to search for attractions near the user.
8	Error and state handling	Loading indicators; messages for no results, connection issues, or API limit reached (e.g. Geoapify daily quota).

4. Wireframes

Recovered from v1 (unchanged layout, now backed by Geoapify + Wikipedia + Esri instead of Nominatim + OpenTripMap + OSM tiles).

View 1 - Home and results (desktop and mobile)

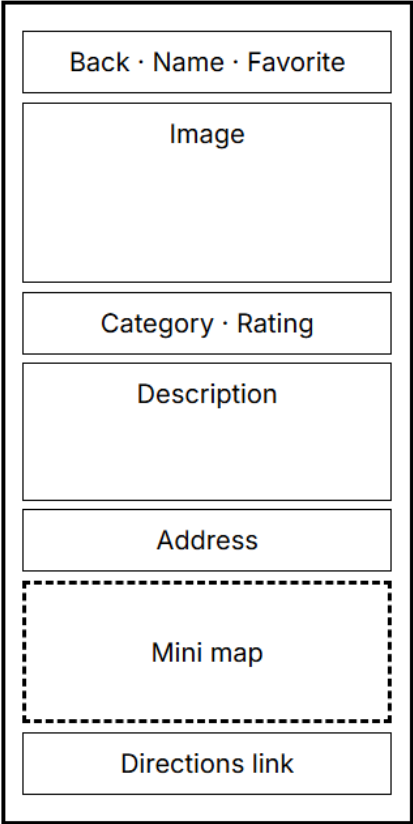
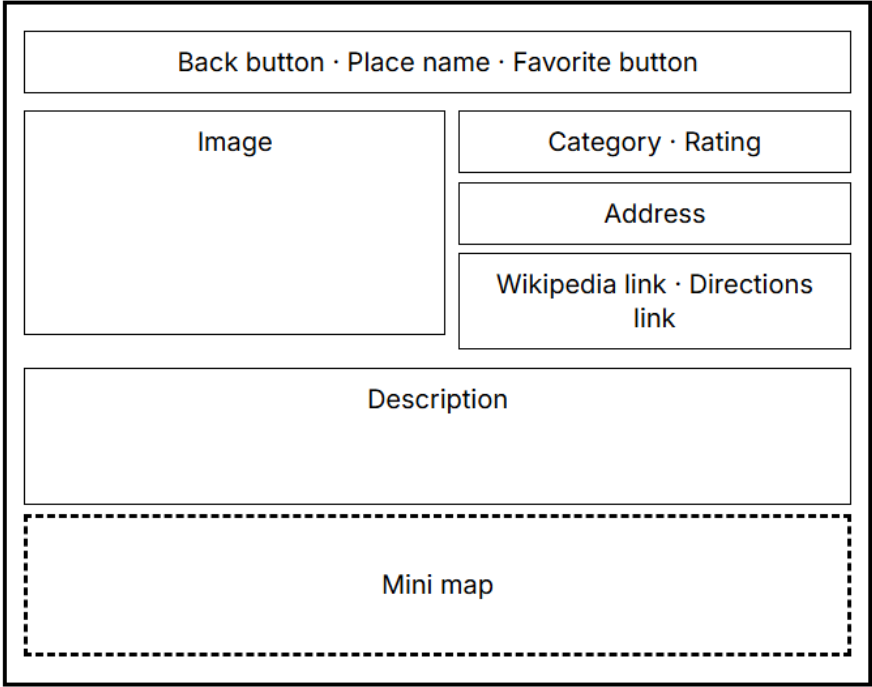
View 1 – Home and results (desktop and mobile)



View 1: Home and results wireframe

View 2 - Place details

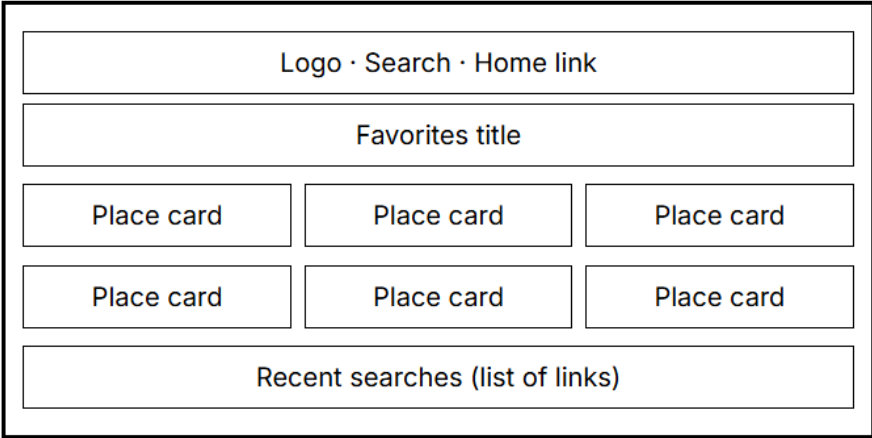
View 2 – Place details



View 2: Place details wireframe

View 3 - Favorites and history

View 3 – Favorites and history



View 3: Favorites and history wireframe

5. External Data

Source	Purpose	URL	Requires API key?
Geoapify Geocoding	Geocoding: converts a typed city or address into latitude/longitude and returns the formatted address. Free plan: 3000 requests/day.	<code>`https://api.geoapify.com/v1/geocode/search`</code> (docs: <code>`https://apidocs.geoapify.com/docs/geocoding/forward-geocoding/`</code> , key at <code>`https://myprojects.geoapify.com`</code>)	Yes
Wikipedia API	Retrieves nearby places (<code>`list=geosearch`</code> , 5 km radius) and the details of each one (<code>`prop=pageimages`</code>)	<code>extracts\</code>	coordinates: image, summary, Wikipedia link).
Esri World Street Map	Map tiles (<code>`.../MapServer/tile/{z}/{y}/{x}`</code>). Free, no key; attribution required.	<code>`https://server.arcgisonline.com/ArcGIS/rest/services/World_Street_Map/MapServer/tile/{z}/{y}/{x}`</code>	No
Leaflet	Map rendering and markers (a mapping library, not a data API).	<code>`https://leafletjs.com/`</code>	No

Removed: OpenStreetMap Nominatim + OSM tiles (blocked/unusable), OpenTripMap (dead/scam docs), CARTO basemaps (now requires API key), Open-Meteo (replaced by Geoapify per key requirement).

Data to store (localStorage):

- ``favorites``: array of saved places (pageid, name, category, coordinates, image).
- Geoapify key: hardcoded in ``js/api.mjs`` (educational project, public teaching key).

6. Module List

Module	Responsibility
<code>`api.mjs`</code>	Geoapify geocode request, Wikipedia geosearch + details requests, and error handling (quota/empty results). No <code>`xid`</code> logic anymore.
<code>`map.mjs`</code>	Initializes Leaflet with Esri tiles (<code>`{z}/{y}/{x}`</code> order), adds/removes markers, syncs the map with the list.
<code>`ui.mjs`</code>	Renders cards, detail modal, filter chips, loaders, and messages. Computes distance (haversine) locally.
<code>`storage.mjs`</code>	Reads/writes favorites in localStorage (keyed by Wikipedia <code>`pageid`</code>).
<code>`main.mjs`</code>	Entry point: wires up events, handles geolocation, and switches between views.

7. Graphic Identity

(unchanged - see v1 PDF)

- Primary Teal ``#0F6B6B``, accent Orange ``#E8743B``, background White ``#FFFFFF``, text Dark gray ``#1F2A2E``.
- Inter 400/500, fallback system-ui, sans-serif.
- Cards 10 px rounded corners; orange CTAs; teal active chips; heart icon for favorites.

8. Timeline (Weeks 5-7)

Week	Focus	Deliverables
5	Foundation and data	Project structure and repository; basic responsive HTML/CSS; Geoapify API key ; city search with Geoapify; list of attractions from Wikipedia Geosearch.
6	Map and interaction	Leaflet map with Esri tiles + markers synced to the list; category filter; place detail view (Wikipedia image/summary); loading and error messages.
7	Persistence and polish	Favorites with localStorage (by `pageid`) and favorites view; "Use my location" button; final styling and basic accessibility; testing on mobile and desktop; deployment on GitHub Pages; final presentation.

9. Project Planning (Trello)

Board link: <https://trello.com/b/bGgvGUAd/places-visit-guide>

To Do (updated):

- [x] Get **Geoapify** API key (replaces OpenTripMap/Open-Meteo) - obtained, hardcoded in `js/api.mjs`
- Test endpoints (Geoapify geocode, Wikipedia geosearch+details, Esri tiles)
- Create repository and folder structure
- Define color palette and typography in CSS
- Leaflet map + Esri markers
- Category filter (Wikipedia-based)
- Place detail view (Wikipedia image/summary)
- Favorites with localStorage (by `pageid`)
- Geolocation
- Mobile testing
- Deploy to GitHub Pages

Bugs / Improvements

- (To be filled in during development)

10. Challenges

- **API limits and keys:** Geoapify free plan is 3000 requests/day with API key (hardcoded in `js/api.mjs` - educational project). Wikipedia and Esri need no key but have fair-use limits: search only runs on Enter/button click and shows clear messages if a limit is reached.
- **Tile URL order:** Esri uses `{z}/{y}/{x}` (not OSM's `{z}/{x}/{y}`) - wrong order shows wrong map areas.
- **Incomplete data:** many Wikipedia places lack images or summaries, so default values and layouts that do not break are required (same as v1).
- **Syncing map and list:** keeping markers, cards, and filters consistent when the search or filter changes.
- **Asynchronous flow and errors:** chaining two APIs (geocode first, then Wikipedia geosearch) with async/await and handling network failures.